

ABC-Emo: Belief-Catalyzed Emotional Reasoning for Multimodal Emotion Recognition in Conversation

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Abstract

Multimodal Emotion Recognition in Conversations (MERC) requires a sophisticated synthesis of linguistic, visual, and acoustic signals. Traditional methods predominantly rely on a direct mapping from activating events to emotional labels ($A \rightarrow C$), often overlooking the internal cognitive appraisal process. Consequently, these "black-box" approaches struggle to distinguish surface-level stimuli from belief-driven emotional transitions. To address this, we reformulate MERC as a belief-mediated reasoning task and propose ABC-Emo, a framework inspired by the ABC emotion theory. Specifically, ABC-Emo mediates the transition from Activating Events (A) to emotional Consequences (C) by explicitly integrating vocal, facial, and contextual Beliefs (B). To emulate human cognitive flexibility, we introduce a Dynamic Belief Update (DBU) module that calibrates historical context with real-time perceptual cues via a gated cross-modal mechanism. By synergizing multi-source beliefs with refined contextual states, ABC-Emo consolidates evidence into a unified sequence for joint reasoning via Large Language Models (LLMs). Extensive experiments on AvaMERG, MELD, and IEMOCAP benchmarks demonstrate that ABC-Emo consistently outperforms state-of-the-art methods. Qualitative analyses further confirm that our belief-catalyzed approach provides a traceable and robust reasoning path, particularly in complex scenarios involving expectation violations. Code is available at <https://github.com/x1bhm/ABC-Emo>.

CCS Concepts

• **Applied computing** → **Psychology**; • **Information systems** → **Multimedia information systems**.

Keywords

Multimodal emotion recognition in Conversation, Belief Modeling, Large Language Models, ABC Emotion Theory

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1 Introduction

Multimodal Emotion Recognition in Conversations (MERC) [8, 37, 41, 46] is a cornerstone of affective computing, aiming to infer a speaker's emotional state by synthesizing textual, vocal, and facial signals. Beyond simple categorization, emotion represents a sophisticated cognitive process that shapes how individuals interpret events and formulate social expectations. Consequently, achieving natural and empathetic Human-Computer Interaction (HCI) [10, 33, 44] demands more than surface-level pattern recognition; it requires a deep understanding of the cognitive mechanisms that drive emotional responses during social exchange. While traditional paradigms have significantly progressed by leveraging recurrent frameworks [20, 37] to model temporal dependencies, graph-based architectures [17, 24, 27, 43] to capture complex inter-speaker interactions, or attention-driven mechanisms [28, 35, 38] to optimize temporal modeling, and more recently, Large Language Models (LLMs) [2, 7, 25, 47, 55, 58, 62] to enhance semantic extraction, these advancements remain largely architectural in nature.

Despite such technical refinements, existing methodologies suffer from a fundamental conceptual limitation: they are anchored in an event-driven formulation that directly maps multimodal observations to emotion labels. This direct mapping ($A \rightarrow C$) implicitly treats emotions as spontaneous reactions to external stimuli, largely bypassing the internal cognitive appraisal process that mediates between activating events and affective consequences [39]. Consequently, these "black-box" approaches fail to account for the dynamic evolution of a speaker's internal beliefs—the true causal drivers of emotional shifts. This gap becomes particularly evident when an emotional response is triggered by expectation violation rather than an event's surface-level valence, leaving the underlying cognitive logic of the emotion unaddressed, as illustrated in Fig. 1.



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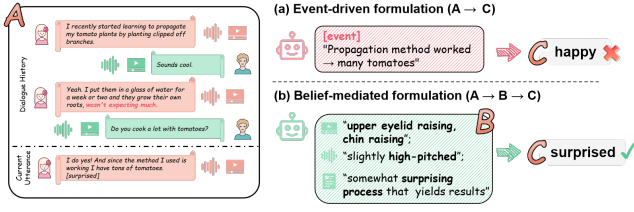


Figure 1: Comparison of Reasoning Paradigms. (a) Event-driven models ($A \rightarrow C$) typically struggle with affective dissonance, where textual events contradict the speaker’s actual non-verbal expressions. (b) In contrast, our Belief-mediated approach ($A \rightarrow B \rightarrow C$) introduces a structured Belief System (B) as a cognitive mediator. By synergizing Perceptual Beliefs (e.g., facial and vocal nuances) with Contextual Beliefs (e.g., expectation violations), this system resolves emotional ambiguities by aligning external stimuli with the speaker’s internal mental activity.

Drawing on the ABC emotion theory proposed by Albert Ellis [14], we argue that emotional consequences (C) are not deterministic outputs of activating events (A); rather, they are fundamentally mediated by an individual’s internal beliefs and cognitive interpretations (B). This paradigm shift suggests that emotions emerge from how an event is interpreted relative to one’s prior expectations. Accordingly, we identify two critical, yet underexplored, methodological challenges: 1) The absence of belief-mediated appraisal, where models lack the cognitive “filter” required to interpret situational nuances; and 2) The static representation of contextual priors, which fails to account for how belief states are dynamically revised by immediate, real-time multimodal evidence.

In this paper, we reformulate MERC as a belief-mediated emotional reasoning task and propose ABC-Emo, a cognitively-inspired framework for belief-catalyzed emotional reasoning. Departing from the conventional *Event* $\rightarrow$ *Emotion* mapping, we introduce an Affect-aware Belief Generation (ABG) module. This module explicitly operationalizes “Belief” (B) as a cognitive mediator, bridging external perceptual stimuli with internal semantic interpretation via belief-related cues. Furthermore, we develop a Dynamic Belief Updating (DBU) mechanism to perform soft multimodal fusion, continuously refining contextual priors with real-time multimodal evidence. By establishing this reasoning path, ABC-Emo transcends simplistic pattern matching and enables more robust emotional reasoning. Moreover, it provides a traceable reasoning mechanism that reveals the cognitive process underlying emotion prediction, particularly in complex scenarios involving expectation violations.

The main contributions of this paper are summarized as follows:

- We propose **ABC-Emo**, a framework that explicitly operationalizes Belief (B) as a cognitive mediator through hierarchical reasoning strategies, effectively transforming black-box classification into a transparent and evidence-based reasoning sequence.
- By refining contextual beliefs with real-time multimodal cues, we develop a dynamic belief updating mechanism that mimics human cognitive flexibility to adapt to sudden emotional shifts and nuanced conversational reversals.

- Evaluations on AvaMERG [61], IEMOCAP [4], and MELD [40] validate the effectiveness, robustness, and interpretability of our cognitively-grounded framework.

2 Related Work

2.1 Multimodal Emotion Recognition in Conversations

Multimodal Emotion Recognition in Conversations (MERC) aims to infer speakers’ emotional states from multimodal dialogue signals (text, audio, and visual). Existing approaches can be broadly grouped into three categories. **Sequence-based methods** model conversations as temporal sequences to capture contextual dynamics, using recurrent or Transformer-based architectures such as DialogueRNN [37], DialogueTRM [38], and HiTrans [28]. **Graph-based methods** encode dialogues as structured graphs to model complex speaker interactions and long-range dependencies, including DialogueGCN [17], MMGCN [24], and MM-DFN [21]. **LLM-based methods** leverage large pre-trained language models to enhance multimodal understanding via instruction-following and reasoning [3, 7, 32, 57, 61]. In addition, external commonsense knowledge has been incorporated to improve contextual reasoning [42, 49, 50]. Despite their architectural diversity, these methods predominantly emphasize representation learning, cross-modal fusion, or generative capability, and formulate emotion recognition as a direct mapping from multimodal inputs to emotion labels. Such paradigms largely overlook the intermediate cognitive processes through which emotions are formed, limiting their ability to model how speakers and dynamically adjust their affective states.

2.2 MLLM-based Emotion Reasoning

The rapid evolution of Multimodal Large Language Models (MLLMs) has shifted affective computing from classification toward generative emotion reasoning. Building on vision-language models such as LLaVA [34] and BLIP-2 [29], recent approaches integrate modality-specific encoders (e.g., Video-LLaMA [60]) to align visual and acoustic signals with LLMs, interpreting non-verbal cues [36] and generating affective states. Leveraging long-context capabilities, conversational frameworks like InstructERC [26] reformulate emotion recognition into generative tasks, while prompt-based methods [6, 16, 19, 30, 51, 52, 54, 63] elicit structured rationales to improve interpretability. Parallely, new benchmarks have been introduced to evaluate MLLM-based emotion reasoning [2, 25]. Building on these advances, we extend MLLM-based emotion reasoning with cognitive grounding, explicitly modeling dynamic interactions between multimodal cues and internal cognitive states for fine-grained, interpretable understanding.

2.3 Cognitively-Inspired Emotion Modeling

Cognitively-inspired approaches move beyond surface signal modeling by approximating human appraisal processes. Recent studies integrate emotion, causal reasoning, and commonsense knowledge into unified frameworks. For instance, ECC [53] models their interactions via a collaborative framework, enabling coherent and interpretable reasoning. Beyond text, such modeling extends to

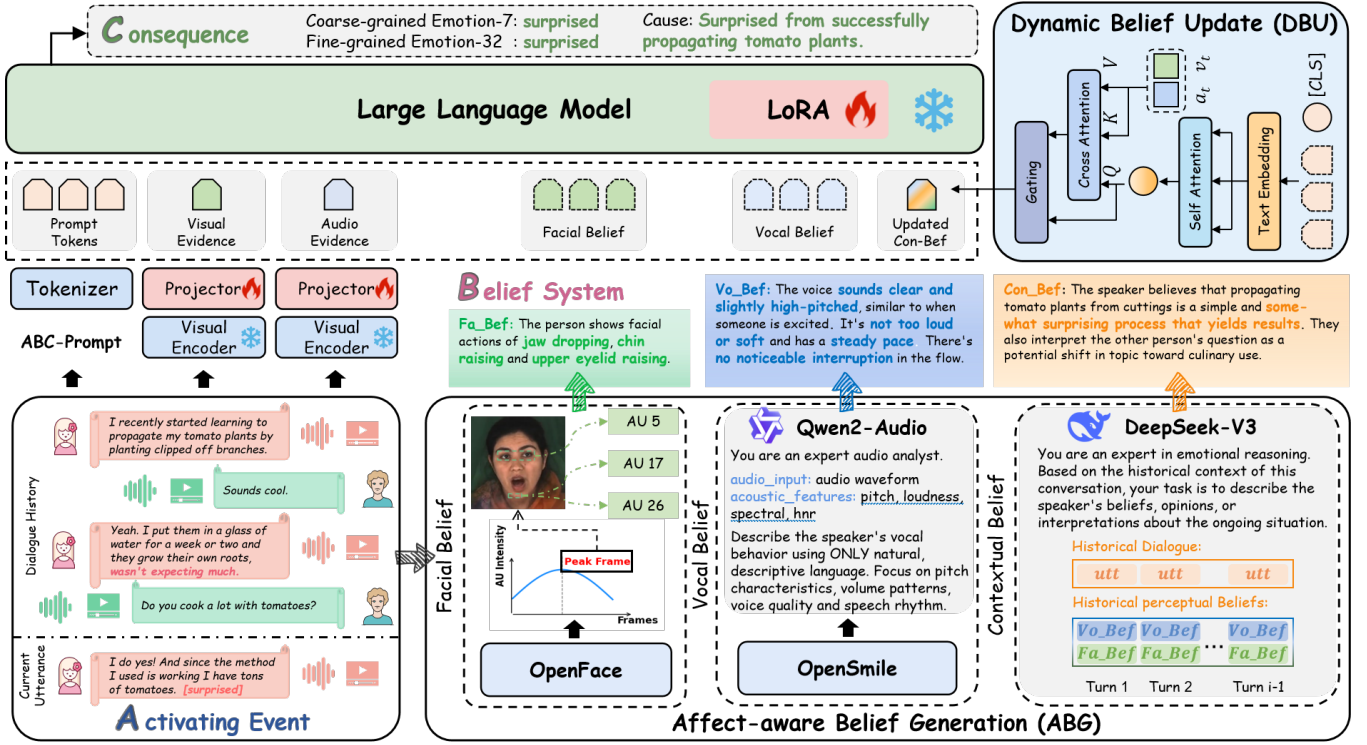


Figure 2: Overview of the proposed ABC-Emo framework. Our model reformulates emotion recognition as a belief-mediated reasoning process.

physiological signals [12] and neural representations [56], capturing long-term dependencies and dynamic causal interactions. With the rise of LLMs, cognitively grounded reasoning has further advanced via multi-step inference frameworks and dual-process cognitive mechanisms [31]. In parallel, Ellis’s ABC theory [14] provides a principled foundation by decomposing emotion formation into activating events (A), belief systems (B), and emotional consequences (C). This emphasizes that emotions stem from subjective interpretations rather than external stimuli, motivating belief-centered modeling in multimodal emotion reasoning.

3 Methodology

3.1 Problem Formulation

ABC-based Task Reformulation. Inspired by the ABC emotion theory [14], we reformulate this task as a structured Belief-Mediated Emotional Reasoning process. For a given conversational turn t , the task is to infer the emotional consequence C_t based on the activating event A_t (current utterance T_t and dialog contexts) and the internal cognitive belief B_t (internal mental activities). We represent each dialogue state as a triplet $S_t = (A_t, B_t, C_t)$:

- **Activating Event (A_t):** The objective input consisting of the current utterance T_t , dialogue history $H_{<t}$, and raw multimodal signals Vid_t, Aud_t .

- **Belief System (B_t):** The subjective cognitive state, formulated as $B_t = \{B_v^t, B_f^t, \tilde{B}_c^t\}$. Here, B_v^t and B_f^t denote explicit vocal and facial beliefs derived from perception, while $\tilde{B}_c^t$ represents a latent contextual belief refined via a Dynamic Belief Updating (DBU) module.
- **Emotional Consequence (C_t):** The reasoning outcome $C_t = \{E_7^t, E_{32}^t, Y_c^t\}$, where E_7^t and E_{32}^t are coarse- and fine-grained emotional categories, and Y_c^t is the rationalized explanation of the emotion’s cause.

As illustrated in Fig. 2, this $A \rightarrow B \rightarrow C$ logic transforms MERC from a simple pattern-matching task into a transparent reasoning chain, where emotional outcomes are explicitly grounded in the speaker’s internal belief system.

Belief-catalyzed Emotional Reasoning. The framework aims to optimize a reasoning function $f : (A_t, B_t) \rightarrow C_t$, which maximizes the joint probability of predicting emotional labels and their corresponding causal explanations. Specifically, a Large Language Model (LLM) serves as the reasoning engine, processing an integrated input of high-level multimodal dialogue embeddings and affect-aware beliefs to infer the Emotional Consequence (C_t). This reasoning process is guided by a structured prompt that synthesizes: (i) the Activating Event (A_t), derived from the dialogue history $H_{<t}$ and current utterance T_t ; and (ii) the Belief System (B_t), which fuses

perceptual beliefs $\{B_v^t, B_f^t\}$ with the dynamic contextual belief token $\tilde{B}_c^t$. The Emotional Consequence (C_t) is then generated as the reasoning target, encompassing coarse-grained (E_7), fine-grained (E_{32}) emotions, and the rationalized cause (Y_c). A representative prompt structure is as follows:

“You are an expert in emotional reasoning... Conversation context: $\{H_{<t}\}$, Current utterance: $\{T_t\}$. Perceptual beliefs: $\{B_{face}, B_{voice}\}$. After integrating perceptual evidence, the updated contextual belief: $\langle belief \rangle$. Predict coarse-grained $\langle E_7 \rangle$, fine-grained $\langle E_{32} \rangle$, and infer the cause $\langle Y_c \rangle$.”

Emotion Cause Annotation. Existing multimodal conversation datasets (e.g., IEMOCAP [4], MELD [40], AvaMERG [61]) primarily provide categorical emotion labels but lack explicit emotion cause (Y_c) annotations. To address this limitation and support our belief-mediated reasoning framework, we utilize DeepSeek-V3 [11] to augment these datasets with cause labels Y_c and infer listener emotions for AvaMERG. Detailed procedures are in Appendix A.

3.2 Affect-aware Belief Generation

The belief system B_t is constructed by integrating multi-source information to form a holistic understanding of the speaker’s internal state.

Facial Belief. We leverage facial actions as a visual internal beliefs. After extracting raw Action Units (AUs) via OpenFace [1], we implement an activation frequency filter to ensure the robustness of the visual evidence. Specifically, to filter out transient noise, we only retain AUs that remain active in over 80% of the video frames, effectively suppressing sporadic outliers. For each frame f_i , the resulting stably activated set S_{f_i} is used to compute the total intensity I_{f_i} . The frame exhibiting the highest aggregate intensity is then selected as the peak expressive frame f^* , serving as the primary visual stimulus for subsequent reasoning,

$$I_{f_i} = \sum_{a \in S_{f_i}} I_{f_i,a}, \quad f^* = \arg \max_{1 \leq i \leq N} I_{f_i}, \quad (1)$$

where $I_{f_i,a}$ is the intensity of a -th AU in frame f_i , N is the total number of frames. we select the top- k AUs with the highest aggregate intensities to serve as the core visual evidence. To transform these low-level physical signals into high-level Visual Beliefs (B_f), we employ a mapping function $\mathcal{M}_{\text{FACS}}(\cdot)$ grounded in the Facial Action Coding System (FACS) [13]. Specifically, for each selected $AU_{f^*}^k$, the function retrieves its corresponding anatomical description (e.g., “Inner Brow Raiser”, “Chin Raiser”, “Lip Corner Puller”):

$$\text{Desc}_k = \mathcal{M}_{\text{FACS}}(AU_{f^*}^k), \quad (2)$$

The final facial belief B_f^t is formulated as:

“The person shows facial actions of $[Desc_1], [Desc_2], \dots, [Desc_k]$.”

Vocal Belief. To generate the Vocal Belief (B_v^t), we employ Qwen2-Audio-Instruct [9] as the central processing engine. By integrating the raw waveform $Audio_t$ with the explicit acoustic intensities extracted via OpenSMILE [15], the model functions as a perceptive “brain” that translates low-level signal fluctuations into high-level semantic descriptions. Specifically, we provide the

model with a structured prompt that reconciles numerical evidence (e.g., pitch and loudness scores) with raw auditory perception. This allows Qwen2-Audio to generate a nuanced, multidimensional appraisal of the speaker’s voice, effectively forming the vocal component of our affect-aware belief system.

Prompt for Vocal Belief Generation

System: You are an expert audio analyst. You receive precise acoustic measurements but describe vocal characteristics using only natural, non-technical language without any numbers or measurements.

Detailed acoustic measurements available:

- Pitch: mean={pitch_mean}, range={pitch_range}
- Loudness: level={loudness}, dynamics={loudness_std}
- Spectral: centroid={spectral_centroid}, flux={spectral_flux}
- HNR: mean={hnr_mean}, std={hnr_std}

Task: Based on the precise measurements provided, describe the speaker’s vocal behavior using ONLY natural, descriptive language.

- Focus on: Pitch height, volume patterns, voice quality, and speech rhythm.

Output Format:

- Start with “The voice” and keep it concise.

Contextual Belief. The core of a belief system is the speaker’s subjective interpretation of events. Building on this, we construct the contextual belief B_c^t , which encapsulates the accumulated cognitive state up to turn t . To ensure that B_c^t reflects a coherent and stable psychological state rather than transient reactions, we incorporate not only the dialogue history but also the historical vocal and facial beliefs ($B_v^{1:t-1}, B_f^{1:t-1}$) as inputs to DeepSeek-V3 [11] ($\mathcal{G}$):

$$B_c^t = \mathcal{G}(\{(u_i, B_v^i, B_f^i)\}_{i=1}^{t-1}). \quad (3)$$

By synthesizing these multimodal histories, we provide the model with a comprehensive view of the speaker’s state, enabling a more robust appraisal of how their internal beliefs evolve over time.

Prompt for Contextual Belief Generation

You are an expert in emotional reasoning. Infer the speaker’s **contextual belief** from historical dialogue and multimodal behaviors. Do NOT describe emotions.

Historical Dialogue: {Dialogue history}

Historical Audio/Video Descriptions:

- Audio cues: {historical vocal beliefs}
- Visual cues: {historical facial beliefs}

Current Utterance: {Current utterance}

Task: What are the speaker’s beliefs and interpretations at this point? Focus on their internal perspective of the situation (1–2 concise sentences).

3.3 Dynamic Belief Update

While the contextual belief B_c^t encodes historical interpretations, emotional states are inherently dynamic and require continuous refinement from immediate perceptual evidence. To refine the speaker’s internal state with real-time audiovisual evidence, we

propose the **Dynamic Belief Update (DBU)** module, which performs soft multimodal fusion to dynamically update the contextual belief.

Contextual belief $\mathbf{B}_c^t$ is first projected into a d -dimensional embedding space $\mathbf{C}^t \in \mathbb{R}^{L \times d}$. A learnable [CLS] token $\mathbf{c}_{cls}^t$ is prepended to capture global dependencies via Multi-Head Self-Attention (MHSA):

$$\mathbf{h}_c^t = \text{LN}(\text{MHSA}(\mathbf{X}^{(t)} + \mathbf{X}^{(t)})_{[cls]}) \quad (4)$$

where $\mathbf{h}_c^t \in \mathbb{R}^{1 \times d}$ represents the aggregated historical belief state, $\mathbf{X}^{(t)} = [\mathbf{c}_{cls}^t; \mathbf{C}^t] \in \mathbb{R}^{(L+1) \times d}$, $\text{LN}(\cdot)$ denotes layer normalization.

To incorporate instantaneous stimuli, we employ a Cross-Modal Attention mechanism. Here, $\mathbf{h}_c^t$ serves as the *Query*, while the perceptual embeddings $\mathbf{P}^t \in \mathbb{R}^{2 \times d}$ (comprising audio features $\mathbf{a}_t \in \mathbb{R}^{1 \times d}$ and video features $\mathbf{v}_t \in \mathbb{R}^{1 \times d}$) act as *Keys* and *Values*. The resulting perceptual adjustment $\mathbf{B}_p^t$ captures how real-time evidence modulates the existing belief:

$$\mathbf{B}_p^t = \text{CrossAttn}(\mathbf{h}_c^t, \mathbf{P}^t, \mathbf{P}^t) \quad (5)$$

Finally, we introduce a gated mechanism to adaptively update the prior belief with perceptual cues:

$$\begin{aligned} \mathbf{g} &= \sigma([\mathbf{h}_c^t; \mathbf{B}_p^t]), \\ \tilde{\mathbf{B}}_c^t &= \text{LN}(\mathbf{g} \odot \mathbf{B}_c^t + (1 - \mathbf{g}) \odot \mathbf{h}_c^t) \end{aligned} \quad (6)$$

where $\mathbf{g}$ denotes the update gate, $\sigma(\cdot)$ is sigmoid function, $[]$ is feature concatenation. The update gate $\mathbf{g}$ balances the preservation of historical context ($\mathbf{g} \rightarrow 0$) against the integration of salient physical cues ($\mathbf{g} \rightarrow 1$).

3.4 Belief-catalyzed Emotion Reasoning

To achieve nuanced reasoning, we construct a unified input sequence $\mathbf{I}_t$ by combining two types of evidence: **Hard Prompts** and **Soft Prompts**. The former provides symbolic context, including the task instruction $\mathcal{I}$, dialogue history $H_{<t}$, current text T_t , and the previously constructed semantic beliefs $\{\mathbf{B}_f^t, \mathbf{B}_o^t\}$. The latter introduces **Soft Prompts** as continuous “virtual tokens” that project raw audiovisual features ($\mathbf{a}_t, \mathbf{v}_t$) and the refined contextual belief $\tilde{\mathbf{B}}_c^t$ into the LLM’s embedding space via trainable linear layers $\mathcal{W}$,

$$\mathbf{I}_t = [\text{LLM}(\mathcal{I}, H_{<t}, T_t, \mathbf{B}_f^t, \mathbf{B}_o^t) \oplus \mathcal{W}_a(\mathbf{a}_t) \oplus \mathcal{W}_v(\mathbf{v}_t) \oplus \mathcal{W}_c(\tilde{\mathbf{B}}_c^t)], \quad (7)$$

where $\oplus$ denotes concatenation. By combining explicit symbolic descriptions with implicit feature-level evidence, the reasoning engine can better understand the speaker’s state and yield structured emotional results $\mathbf{I}_t = (E_7^t, E_{32}^t, Y_c^t)$ with higher interpretability.

ABC-structured Prompt for Emotion Reasoning

You are an expert in emotional reasoning. When analyzing emotions, you consider both the situation and how the speaker interprets it through their perceptions and prior context.

Conversation Context: {Dialogue history}

Current Utterance: {Current utterance}

Perceptual Evidence (Latent):

- Visual representation: <Vid> (via $\mathcal{W}_v(\mathbf{v}_t)$)
- Audio representation: <Aud> (via $\mathcal{W}_a(\mathbf{a}_t)$)

Perceptual Beliefs (Semantic):

- Facial Belief: {facial_belief}
- Vocal Belief: {vocal_belief}

Updated Contextual Belief (Latent): <belief> (via $\mathcal{W}_c(\tilde{\mathbf{B}}_c^t)$)

Outputs:

- Coarse-grained emotion: <emotion7>
- Fine-grained emotion: <emotion32>
- Emotion cause: <cause>

3.5 Training Objectives

Emotion Classification and Generation. The model jointly predicts coarse-grained (E_7) and fine-grained (E_{32}) emotions using cross-entropy loss: $\mathcal{L}_{cls} = \mathcal{L}_{E_7} + \mathcal{L}_{E_{32}}$. Concurrently, the emotion cause Y_c is supervised by an generation loss:

$$\mathcal{L}_{gen} = - \sum_i \log P(y_i | y_{<i}, \mathbf{I}_t). \quad (8)$$

Belief Consistency Regularization. To ensure that the updated contextual belief $\tilde{\mathbf{B}}_c^t$ faithfully captures the speaker’s internal state, we introduce a consistency regularizer $\mathcal{L}_{cons}$. Specifically, we pass the latent belief vector $\tilde{\mathbf{B}}_c^t$ through a dedicated linear classifier to derive a belief-based emotion distribution p_{bel} , which represents a coarse-grained emotional mapping of the internal cognitive state. We then employ KL-divergence to minimize the discrepancy between p_{bel} and the primary reasoning engine’s prediction p_{main} :

$$\mathcal{L}_{cons} = D_{KL}(p_{bel} || p_{main}). \quad (9)$$

Overall Objective. The total loss is the sum of the three objectives:

$$\mathcal{L}_{total} = \mathcal{L}_{gen} + \mathcal{L}_{cls} + \mathcal{L}_{cons}. \quad (10)$$

Table 1: Statistics of the evaluated datasets.

Dataset	# Dialogue	# Utterance	# Emotion	# Cause
AvaMERG	28,601	131,580	7 / 32	131,580
MELD	1,433	13,708	7	13,708
IEMOCAP	151	7,380	6	7,380

3.6 Implementation Details

4 Experiments

4.1 Experimental Settings

Datasets. We evaluate our framework on three widely used benchmarks: AvaMERG [61], MELD [40], and IEMOCAP [4], with their detailed statistics summarized in Table 1. To facilitate the proposed belief-mediated reasoning, we augmented these datasets with additional labels for emotion causes (Y_c) and listener perspectives. Considering the relatively smaller scale of MELD and IEMOCAP, we retain their original categorical emotion sets for coarse-grained recognition rather than expanding them to 32-class labels. The detailed annotation procedures are documented in Appendix A.

Implementation Details. We adopt Llama2-7B [48] fine-tuned via LoRA [23] (rank=16) as backbone, with ImageBind [18] for multimodal features. The DBU module (hidden dim 4096, 8 heads) is trained from scratch. We train for 3 epochs using AdamW ($lr = 5 \times 10^{-5}$ with cosine decay, batch size 8) on a single NVIDIA RTX 6000 GPU via DeepSpeed ZeRO-2.

Table 2: Emotion classification results on AvaMERG dataset for 7- and 32-class emotion recognition. Best results are in bold.

Method	7-class				32-class			
	Acc	UA	Macro-F1	Weighted-F1	Acc	UA	Macro-F1	Weighted-F1
DialogueRNN [37]	86.99	77.52	79.50	86.84	60.78	47.15	47.29	59.82
DialogueCRN [22]	87.24	75.53	79.72	87.08	62.20	48.44	49.38	61.59
MM-DFN [21]	86.33	76.28	78.55	86.02	56.97	40.56	40.35	57.23
M3NET [5]	85.54	74.20	76.65	85.39	59.01	46.55	46.05	58.43
MultiEMO [45]	85.34	79.57	76.91	85.56	56.11	50.89	47.75	57.54
HAUCL [59]	84.11	67.78	71.99	83.73	55.06	39.91	40.41	54.13
Emotion-LLaMA [7]	89.50	81.73	83.30	89.42	72.29	60.93	61.85	71.79
AffectGPT [32]	90.67	82.67	84.45	90.55	74.53	64.12	64.75	73.77
Empatheia [61]	86.75	74.71	77.53	86.51	71.91	61.53	62.29	71.83
ABC-Emo (Ours)	91.54	85.85	86.59	91.48	76.05	67.84	68.50	75.80

Table 3: Emotion classification results (F1-score per class) on MELD and IEMOCAP. Best results are in bold.

Method	MELD (F1-score per class)									IEMOCAP (F1-score per class)							
	Neu.	Sur.	Fear	Sad	Joy	Dis.	Ang.	Acc	w-F1	Hap.	Sad	Neu.	Ang.	Exc.	Fru.	Acc	w-F1
DialogueRNN [37]	75.38	48.10	0.00	28.49	51.23	0.00	45.71	59.35	57.65	31.93	77.42	54.10	64.63	67.86	60.26	60.87	60.74
DialogueCRN [22]	75.76	43.76	6.45	21.92	50.32	0.00	44.29	58.89	56.64	48.76	81.08	60.88	65.33	74.17	58.66	65.31	65.25
MM-DFN [21]	77.25	48.75	0.00	26.70	54.94	0.00	42.86	61.07	58.68	39.37	78.97	64.95	70.72	72.24	65.01	68.45	68.04
M3NET [5]	79.09	59.09	12.12	44.50	62.52	29.37	52.71	66.17	65.56	56.62	77.23	68.89	68.41	75.76	63.30	69.07	68.96
MultiEMO [45]	79.54	58.45	17.65	41.21	65.31	23.91	54.86	67.24	66.12	51.13	82.13	69.04	68.21	74.09	64.06	68.89	69.10
HAUCL [59]	79.23	57.49	26.19	40.62	61.87	31.37	50.38	66.51	65.06	41.38	74.27	62.30	66.11	71.09	63.99	64.70	64.67
Emotion-LLaMA [7]	77.74	53.33	13.51	34.50	58.26	19.05	49.66	64.02	62.19	31.68	70.59	60.12	55.19	66.87	54.73	59.43	58.66
AffectGPT [32]	78.40	53.96	5.63	38.16	57.22	22.22	48.31	63.75	62.46	55.93	73.03	64.11	64.46	63.49	67.38	65.23	65.43
Empatheia [61]	78.39	61.09	33.68	44.82	63.78	37.10	55.31	66.92	66.64	48.40	80.09	66.50	68.33	74.96	66.32	68.50	68.67
ABC-Emo (Ours)	80.27	63.27	31.17	43.23	64.60	37.61	58.01	69.20	68.08	57.32	82.33	71.29	67.74	72.89	69.82	70.97	71.30

Baselines. We evaluate **ABC-Emo** against representative MERC methods, spanning sequence-based architectures like **DialogueRNN** [37] and the cognitive-driven **DialogueCRN** [22]. We also compare with advanced representation alignment and fusion frameworks, including **HAUCL** [59], **MM-DFN** [21], **M3NET** [5], and **MultiEMO** [45]. Finally, we benchmark against the latest LLM-based reasoning paradigms, such as **Emotion-LLaMA** [7], **AffectGPT** [32], and the explainable reasoning model **Empatheia** [61].

4.2 Main Results

We evaluate ABC-Emo on AvaMERG, MELD, and IEMOCAP, with results detailed in Tables 2, 3, and 4. The empirical evidence leads to the following key observations:

Emotion Classification. ABC-Emo establishes new benchmarks across all three datasets, validating the efficacy of belief-catalyzed reasoning. On AvaMERG (Table 2), it yields a **4.79%** accuracy gain over Empatheia and surpasses AffectGPT by **3.75%** in 32-class Macro-F1. This substantial improvement in fine-grained tasks proves that decomposing stimuli into structured beliefs is vital for capturing nuanced affective states. Furthermore, ABC-Emo demonstrates superior robustness on MELD (**68.08%**) and IEMOCAP (**71.30%**). The significant performance gains in traditionally

"confusable" categories, such as *Angry* and *Frustrated*, underscore the framework's capacity to leverage contextual beliefs to resolve subtle cues that often confound traditional methods.

Emotion-Cause Generation. As shown in Table 4, ABC-Emo consistently outperforms baselines across both lexical and semantic dimensions. Notably, it achieves the highest BERTScore across all benchmarks (e.g., **70.73** on AvaMERG), indicating that generated causes are more semantically congruent with ground-truth appraisals. These results confirm that our ABC-structured prompting and belief-aware reasoning empower the model to transcend surface-level correlations. Instead, it provides interpretable and logically coherent explanations that effectively bridge the gap between observed multimodal stimuli and inferred emotional consequences.

4.3 Ablation Studies

We conduct extensive ablation experiments on AvaMERG to evaluate key components of **ABC-Emo** (corresponding results on MELD and IEMOCAP are provided in Appendix E.5).

Emotion Reasoning Mechanism. As shown in Table 5, removing the Belief System (**w/o ABG**) causes the largest drop (-4.97% and -3.97% weighted F1 for 7/32-class), validating our structured reasoning paradigm. Disabling Dynamic Belief Update (**w/o DBU**)

Table 4: Emotion-cause generation results across all datasets evaluated by ROUGE (R), BLEU (B), and BERTScore. Best results are in bold.

Dataset	Method	R-1	R-2	R-L	B-1	B-2	B-3	B-4	BERTScore
AvaMERG	Emotion-LLaMA [7]	36.17	12.23	35.01	16.33	9.76	6.46	4.46	49.00
	AffectGPT [32]	36.03	17.65	33.35	33.17	23.28	17.15	12.93	70.49
	Empatheia [61]	37.67	22.07	35.32	30.81	22.00	16.26	12.61	70.10
	ABC-Emo	38.21	20.42	35.47	31.79	23.46	18.10	14.38	70.73
MELD	Emotion-LLaMA [7]	35.27	9.66	33.67	31.90	16.75	9.23	5.37	56.32
	AffectGPT [32]	35.94	16.91	32.53	35.07	23.82	16.02	10.89	57.34
	Empatheia [61]	33.26	15.80	31.61	31.01	20.61	14.57	10.25	63.06
	ABC-Emo	36.21	17.89	33.76	34.41	24.33	16.72	11.30	66.10
IEMOCAP	Emotion-LLaMA [7]	30.38	11.51	29.59	20.86	14.18	7.93	3.71	53.21
	AffectGPT [32]	31.98	16.15	30.17	29.83	21.31	15.78	12.04	64.51
	Empatheia [61]	30.00	14.80	28.79	26.94	19.13	13.59	9.24	60.47
	ABC-Emo	32.85	15.79	30.94	30.62	21.48	14.60	9.70	65.32

Table 5: Impact of Reasoning Mechanisms.

Variant	Emotion (w-F1)		Cause Generation		
	7-cl	32-cl	R-L	B-4	BScore
ABC-Emo (Full)	91.48	75.80	35.47	14.38	70.73
w/o ABG	86.51	71.83	35.32	12.61	70.10
w/o DBU	90.21	74.49	31.38	10.87	68.59
w/o L_{cons}	91.56	75.42	31.39	13.06	68.70

degrades cause generation (ROUGE-L -4.09%, BLEU-4 -3.51%), confirming the importance of belief calibration. Omitting consistency loss (**w/o** L_{cons}) reduces generation quality (ROUGE-L drops to 31.39), showing its role in aligning reasoning with explanations.

Table 6: Impact of the Belief System. “C” and “P” denote Contextual and Perceptual beliefs respectively.

Variant	Emotion (w-F1)		Cause Generation		
	7-cl	32-cl	R-L	B-4	BScore
ABC-Emo (Full)	91.48	75.80	35.47	14.38	70.73
w/o Contextual (C)	88.62	73.74	34.48	14.10	70.64
w/o Perceptual (P)	91.38	74.84	35.62	13.94	69.47

Belief Construction. Table 6 dissects the belief components. Removing contextual belief (**w/o Contextual (C)**) harms classification (-2.86% for 7-class), indicating that historical interpretation is vital for emotion recognition. While removing perceptual belief (**w/o Perceptual (P)**) slightly improves ROUGE-L, it degrades the overall BERTScore and fine-grained F1, suggesting the full belief system provides the best semantic balance.

Multimodal Feature Representation. In Table 7, using only textual descriptions without latent features (**w/o Latent Features**) severely degrades generation (ROUGE-L -6.44), highlighting the importance of dense multimodal embeddings. Conversely, removing textual descriptions (**w/o Text Descripts.**) shows a decline in

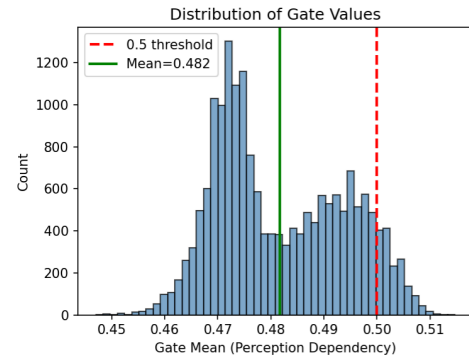
Table 7: Ablation of Feature Representations.

Variant	Emotion (w-F1)		Cause Generation		
	7-cl	32-cl	R-L	B-4	BScore
ABC-Emo (Full)	91.48	75.80	35.47	14.38	70.73
w/o Latent Features	88.55	73.79	29.03	9.95	67.02
w/o Text Descripts.	91.38	74.84	35.62	13.94	69.47

BERTScore, indicating that textual beliefs provide complementary semantic guidance.

4.4 Visualization

Modality Gating Analysis. To interpret the **DBU module**’s internal reasoning, we analyze the gating values (g) mediating **Perceptual Evidence** and **Contextual Beliefs**. Fig. 3 reveals a continuous and balanced distribution centered around 0.482, indicating a soft multi-modal fusion mechanism that dynamically modulates the trade-off between historical contextual priors and real-time sensory inputs.

**Figure 3: Distribution of Gating Values (g) in DBU module. The bimodal pattern reveals an adaptive switch between belief consolidation and perception-driven revision.**

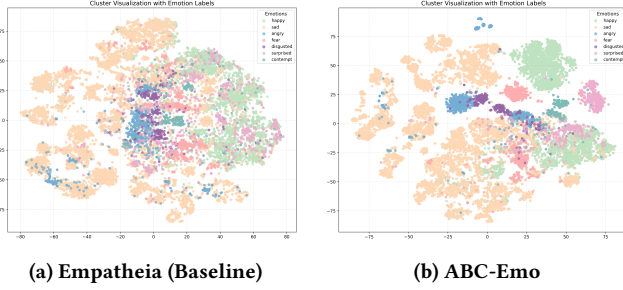


Figure 4: t-SNE visualization of <emotion7> token embeddings on the test set. Compared to the baseline (a), our ABC-Emo (b) demonstrates tighter intra-class clustering and more distinct inter-class separations across different emotions.

Latent Space Visualization. Figure 4 presents the t-SNE visualizations of <emotion7> token embeddings. Compared to the baseline, ABC-Emo produces tighter intra-class clusters and more distinct inter-class boundaries, particularly between visually similar pairs like *happy* (green) and *surprised* (pink). This confirms that belief-guided reasoning effectively structures the emotional representation space, enhancing the discriminability of latent affective features.

Confusion Matrix Analysis. We evaluate the belief system’s impact using row-normalized confusion matrices (Fig. 5), with more comprehensive results provided in Appendix E.4. Compared to the *w/o Belief System* variant, ABC-Emo markedly reduces misclassifications in confusable pairs; e.g., the error of misidentifying *Surprised* as *Happy* drops from 0.25 to 0.11. Furthermore, our architecture significantly improves minority class recognition, with accuracy gains for *Fear* (0.18), *Surprised* (0.24), and *Contempt* (0.14). These results confirm that high-level cognitive appraisals effectively resolve subtle nuances and data imbalance issues that surface-level features fail to capture.

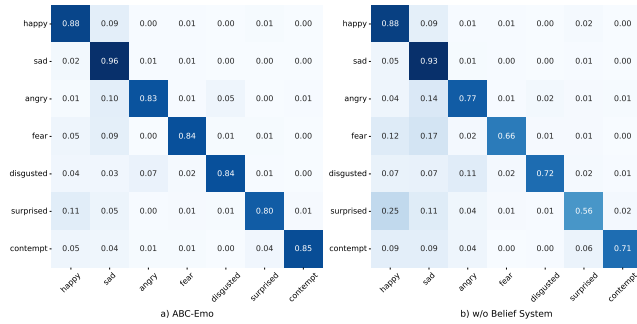


Figure 5: Row-normalized confusion matrices of ABC-Emo and its ablated variants.

4.5 Case Study

To qualitatively interpret ABC-Emo, we analyze a representative case (Fig. 6), with further examples in Appendix F. Here, the baseline (*Empatheia*) is biased by the outcome of “harvesting an abundance of tomatoes” and incorrectly predicts *Happy*, overlooking the

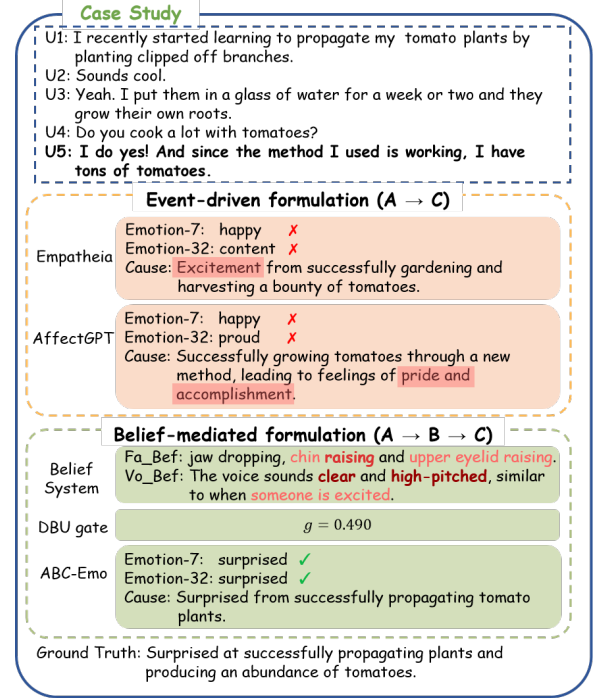


Figure 6: Qualitative analysis of ABC-Emo. ABC-Emo mediates reasoning from Activating events (A) to Consequences (C) via Beliefs (B). Red highlights (token-level saliency) show the model anchoring on key physical evidence to resolve emotional ambiguities. See Appendix F for more examples.

speaker’s prior lack of expectation. In contrast, ABC-Emo structures the reasoning from **Activating events** (A) to **Consequences** (C) via the **Belief System** (B). By attending to the facial “jaw dropping” and vocal “high-pitched” cues, and mediated by the DBU gate ($g = 0.490$), the model successfully performs a *belief revision* from prior skepticism to *Surprised*. This confirms that ABC-Emo prevents “temporal context neglect” by prioritizing stable cognitive beliefs over the immediate emotional valence of the current event, ensuring a causally coherent reasoning chain.

5 Conclusion

We propose ABC-Emo, a belief-catalyzed framework that structures multimodal emotion reasoning into perceptual extraction, contextual belief construction, and dynamic updating. By decomposing inference into cognitive stages, ABC-Emo synergizes multimodal representation with interpretable, belief-guided reasoning. Evaluations on *AvaMERG*, *MELD*, and *IEMOCAP* demonstrate that our approach consistently outperforms state-of-the-art baselines, particularly in scenarios with conflicting affective signals or linguistic biases. Qualitative analyses confirm that our dynamic gating mechanism achieves superior alignment with human-like evidence, providing more granular and causally coherent predictions. Overall, ABC-Emo advances cognitively grounded emotional intelligence by effectively integrating Large Language Models with multimodal affective reasoning.

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